Matrix Chain Multiplication

Definition:

Matrix Chain Multiplication (MCM) is a Dynamic Programming technique used to find the best order for multiplying a sequence of matrices so that the number of scalar multiplications is minimum.

Algorithm:

1. Read the dimensions of all matrices.
2. Create a DP table dp.
3. Set the cost of multiplying one matrix as 0.
4. Consider chains of length 2, 3, 4, ...
5. Try every possible position k to split the chain.
6. Calculate the multiplication cost.
7. Store the minimum cost in dp[i][j].
8. Return the minimum cost for the complete matrix chain.

Out put:

```
Enter number of matrices: 3
Enter dimensions:
Dimension 1: 10
Dimension 2: 20
Dimension 3: 30
Dimension 4: 40

Minimum multiplication cost: 18000
Execution time: 3.885198384523392e-05 seconds
Time Complexity: O(n^3)
Space Complexity: O(n^2)


...Program finished with exit code 0
Press ENTER to exit console.
```

Complexity:

Time Complexity: $O(n^3)$

Space Complexity: $O(n^2)$

Where n = number of matrices.